

Restaurant Sales Analysis

Data Warehousing & OLAP Analysis Report

1. Problem Statement

Restaurants generate large volumes of transactional sales data every day — orders, items sold, prices, payment methods, and timestamps — but this raw data is rarely organized in a way that supports fast, multi-dimensional analysis. Managers need to be able to answer questions like "which menu categories sell best?", "how do sales trend month over month?", and "which items are frequently ordered together?" quickly and reliably.

This project builds a data warehouse for restaurant sales data using a star schema, performs OLAP-style analysis (roll-up, drill-down, slice, dice, ranking) using SQL, and applies market basket analysis (association rule mining) to uncover item combinations that could support combo-meal recommendations.

2. Dataset Description

The dataset (restaurant_sales_raw.csv) contains 500 transaction-level records with the following columns:

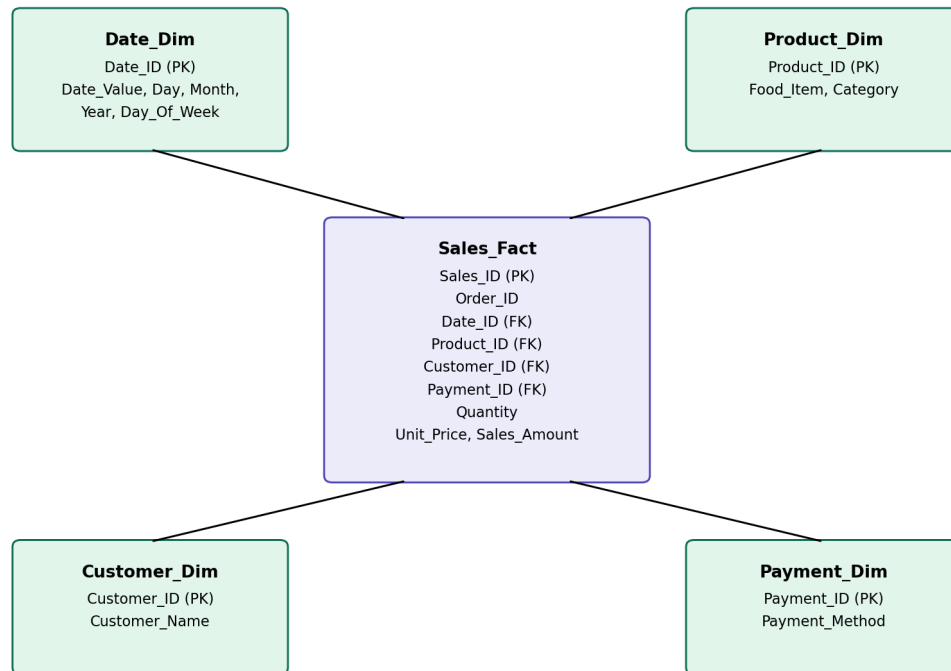
Column	Description
Order ID	Unique identifier for each order
Customer Name	Name of the customer who placed the order
Food Item	Menu item ordered (9 distinct items)
Category	Menu category: Starter, Main, or Dessert
Quantity	Number of units of the item ordered
Price	Unit price at the time of the transaction
Payment Method	Cash, Credit Card, Debit Card, or Online Payment
Order Time	Date and time the order was placed
Sales Amount	Total value of the line item (Quantity x Price)

Data quality checks confirmed no missing values and no duplicate rows across the 500 records.

3. Data Warehouse Schema (Star Schema)

A star schema was designed with one central fact table and four dimension tables, built with an ETL pipeline (described in Section 4). The fact table is kept at the finest grain available — one row per order line item — so it supports both OLAP aggregation and, if needed, basket-level analysis.

Star schema — Restaurant Sales Analysis



Design notes:

- Price varies transaction-to-transaction for the same menu item in this dataset, so Price/Unit_Price is stored as a fact measure, not as a dimension attribute — this keeps Product_Dim clean at 9 rows (one per distinct menu item).
- Surrogate keys (Date_ID, Product_ID, Customer_ID, Payment_ID, Sales_ID) are used to link fact to dimensions, rather than natural keys like item names or raw timestamps.
- This is a star schema (not snowflaked) — all dimensions are flat, single-level tables, which is sufficient for this dataset and keeps OLAP queries simple.

4. ETL Process

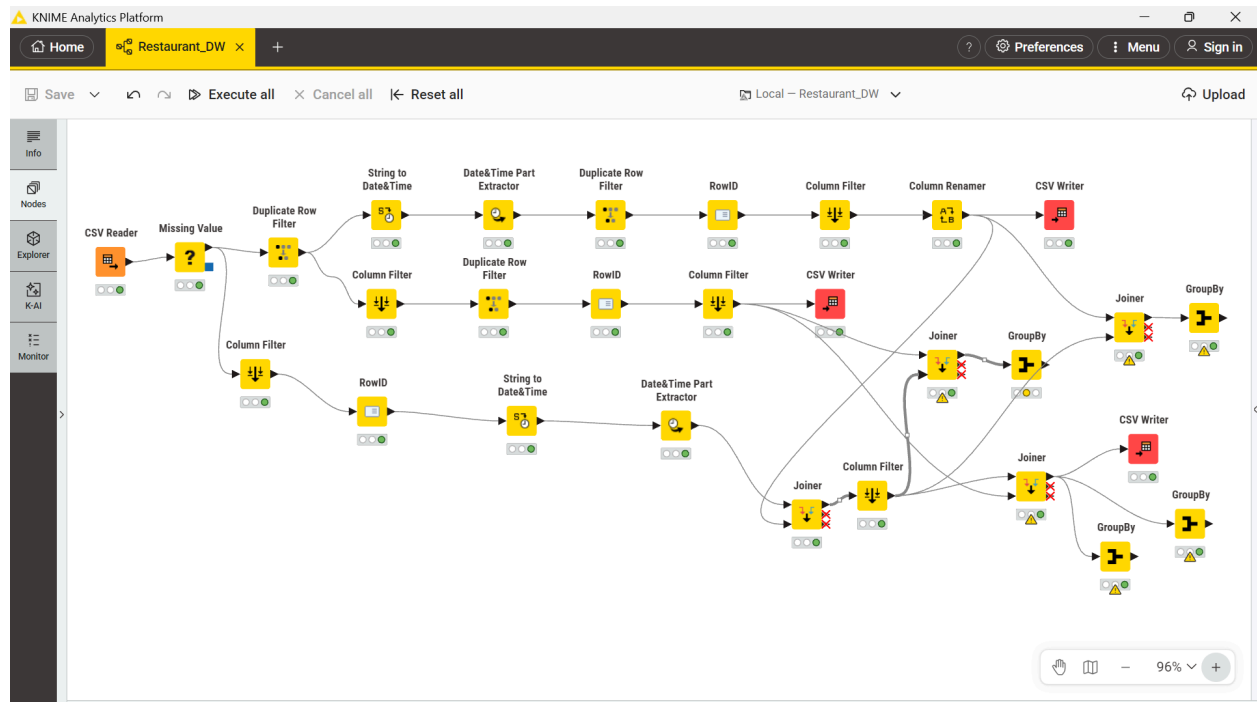
Extract: raw transaction data was read from restaurant_sales_raw.csv.

Transform:

- Checked and confirmed no missing values or duplicate rows
- Parsed Order Time into a proper datetime type and derived Day, Month, Year, and Day-of-week attributes for the Date dimension
- De-duplicated Food Item + Category into a 9-row Product dimension (Price excluded — see Section 3)
- De-duplicated Customer Name into a Customer dimension and Payment Method into a Payment dimension
- Assigned surrogate primary keys to every dimension table

Load: dimension tables were joined back to the transaction data to attach foreign keys, producing the Sales_Fact table, which was loaded into a SQLite database (restaurant_dw.db) as well as exported as CSVs for each table.

An initial version of this ETL pipeline was also built visually in KNIME Analytics Platform (data cleaning -> date/product extraction -> surrogate key assignment -> joins), shown below for reference; the same logic was re-implemented in Python/SQL for this submission to allow direct SQL querying.



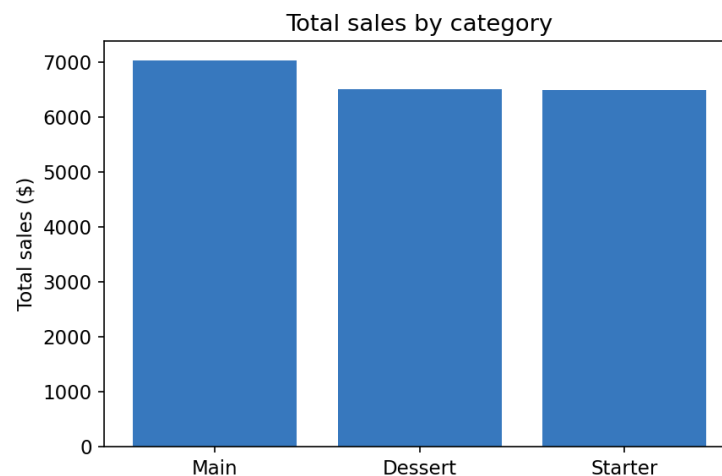
5. SQL Queries and OLAP Analysis

The star schema was loaded into a SQLite database and queried directly with SQL. Five OLAP-style queries were run, covering roll-up, drill-down, slice, dice, and ranking operations.

Query 1 — Roll-up: total sales by category

```
SELECT p.Category, SUM(f.Sales_Amount) AS Total_Sales, SUM(f.Quantity) AS
Total_QtyFROM Sales_Fact f JOIN Product_Dim p ON f.Product_ID = p.Product_IDGROUP BY
p.Category ORDER BY Total_Sales DESC;
```

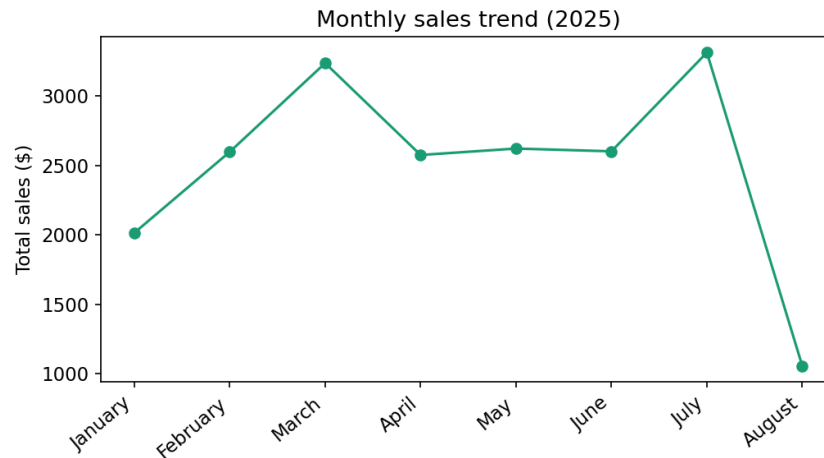
Category	Total Sales	Total Qty
Main	7026.79	523
Dessert	6509.99	504
Starter	6486.36	488



Query 2 — Drill-down: monthly sales trend

```
SELECT d.Year, d.Month, d.Month_Name, SUM(f.Sales_Amount) AS Total_SalesFROM Sales_Fact f JOIN Date_Dim d ON f.Date_ID = d.Date_IDGROUP BY d.Year, d.Month ORDER BY d.Year, d.Month;
```

Month	Total Sales
January	2015.48
February	2598.72
March	3237.43
April	2576.14
May	2622.53
June	2602.60
July	3313.64
August	1056.60



Query 3 — Slice: sales made via Online Payment

```
SELECT p.Food_Item, SUM(f.Sales_Amount) AS Total_Sales, SUM(f.Quantity) AS Total_QtyFROM Sales_Fact fJOIN Product_Dim p ON f.Product_ID = p.Product_IDJOIN Payment_Dim pay ON f.Payment_ID = pay.Payment_IDWHERE pay.Payment_Method = 'Online Payment'GROUP BY p.Food_Item ORDER BY Total_Sales DESC;
```

Food Item	Total Sales	Total Qty
Ice Cream	811.19	66
Pizza	725.38	56
Soup	698.04	45
Brownie	569.97	41
Pasta	564.77	43
Burger	528.23	42
Cake	370.65	36
Fries	328.86	24
Salad	284.66	18

Query 4 — Dice: Main/Dessert sales on weekends

```
SELECT p.Category, d.Day_Of_Week, SUM(f.Sales_Amount) AS Total_SalesFROM Sales_Fact fJOIN Product_Dim p ON f.Product_ID = p.Product_IDJOIN Date_Dim d ON f.Date_ID = d.Date_IDWHERE p.Category IN ('Main', 'Dessert') AND d.Day_Of_Week IN ('Saturday', 'Sunday')GROUP BY p.Category, d.Day_Of_Week ORDER BY p.Category, d.Day_Of_Week;
```

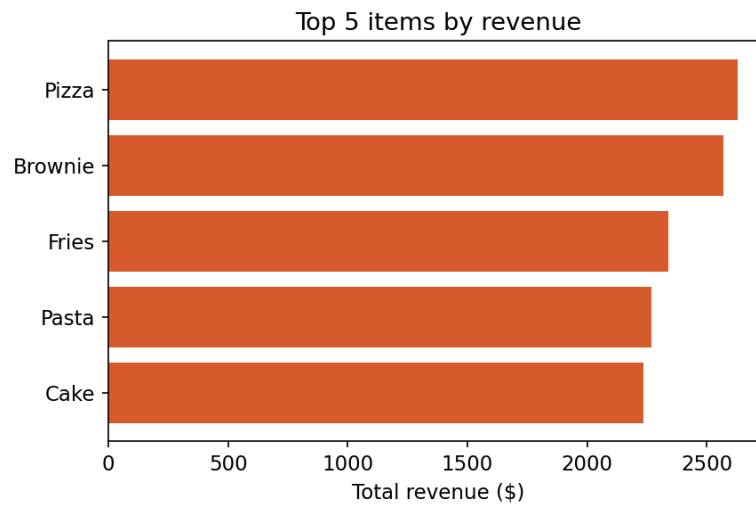
Category	Day	Total Sales
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Dessert	Saturday	958.89
Dessert	Sunday	1194.00
Main	Saturday	636.45
Main	Sunday	1277.97

Query 5 — Ranking: top 5 items by revenue

```
SELECT p.Food_Item, p.Category, SUM(f.Sales_Amount) AS Total_Revenue,
SUM(f.Quantity) AS Total_Qty_Sold, ROUND(AVG(f.Unit_Price),2) AS Avg_Unit_PriceFROM
Sales_Fact f JOIN Product_Dim p ON f.Product_ID = p.Product_IDGROUP BY p.Food_Item,
p.Category ORDER BY Total_Revenue DESC LIMIT 5;
```

Item	Category	Revenue	Qty Sold	Avg Price
Pizza	Main	2627.89	200	12.91
Brownie	Dessert	2570.19	192	13.79
Fries	Starter	2340.48	179	13.35
Pasta	Main	2267.73	154	14.18
Cake	Dessert	2236.58	164	13.06



6. Data Mining Methodology and Algorithm

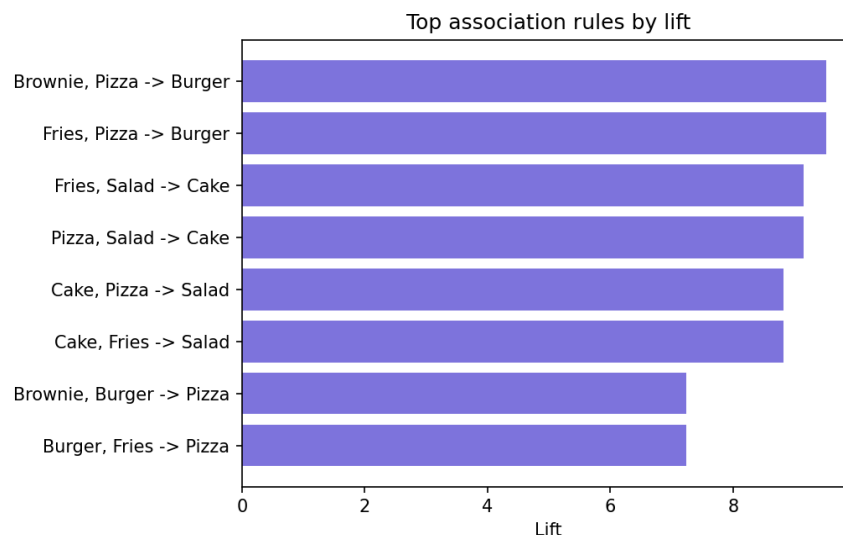
Market basket analysis was performed using the Apriori algorithm (mlxtend library in Python) to find association rules among menu items ordered together. Each order was treated as one transaction; frequent itemsets were mined and converted into rules scored by support, confidence, and lift. Full code and output are in the accompanying Jupyter notebook (DWDM_RESTO_final.ipynb).

Data limitation: only 6 of the 485 distinct orders in this dataset contain 2 or more different items — most customers ordered a single item. This sharply limits the statistical strength of association rule mining here; the minimum support threshold had to be set to the lowest possible value (1/485) to find any multi-item patterns at all. The rules below should be read as early signals from a small sample, not robust patterns.

7. Results with Graphs and Charts

Top association rules by lift:

Antecedent	Consequent	Support	Confidence	Lift
Brownie, Pizza	Burger	0.0021	1.00	9.51
Fries, Pizza	Burger	0.0021	1.00	9.51
Fries, Salad	Cake	0.0021	1.00	9.15
Pizza, Salad	Cake	0.0021	1.00	9.15
Cake, Pizza	Salad	0.0021	1.00	8.82



The category-wise (Query 1), monthly trend (Query 2), and top-item (Query 5) charts from Section 5 are also part of the overall results — together they show that Main dishes lead in total revenue, sales peaked in March and July, and Pizza is the single best-selling item by revenue.

8. Dashboard

The SQL query results and charts above (category-wise sales, monthly trend, top items, and top association rules) form the core content of the dashboard. These were built in Python (pandas + matplotlib) directly from the SQLite star schema, so the same queries can be reused to build an interactive Power BI or Tableau dashboard by connecting directly to restaurant_dw.db or the exported CSV tables.

9. Conclusion and Future Scope

This project took raw restaurant transaction data through a full data warehousing pipeline: cleaning, star-schema design, ETL, SQL-based OLAP analysis, and association rule mining. Key findings: Main dishes generate the most revenue overall, sales peaked in March and July and dipped sharply in August, Pizza and Brownie are the top revenue-driving items, and a small set of items (Pizza, Brownie, Burger, Fries, Salad, Cake) show strong co-occurrence patterns worth exploring for combo deals.

Future scope:

- Collect richer basket-level data (more multi-item orders) to strengthen the association rule mining
- Add a Branch/Restaurant dimension if the business expands to multiple locations
- Build a live, interactive Power BI/Tableau dashboard connected directly to the warehouse
- Extend OLAP analysis with customer segmentation (repeat customers, average spend per customer)